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Breathing apparatus and method of communicating using breathing apparatus

Abstract

A method of communicating using a breathing apparatus includes receiving an audio signal from a sound acquisition unit. The method further includes determining a state of the breathing apparatus based on the received audio signal. The state is at least one of a first state and a second state. The method further includes applying a first filter on the audio signal if the determined state is the first state. The first filter has a first frequency response. The method further includes applying a second filter on the audio signal if the determined state is the second state. The second filter has a second frequency response different from the first frequency response of the first filter. The method further includes generating an output signal based on the application of the first filter or the second filter. The method further includes receiving the output signal at an output device.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to the field of personal protective equipment, such as a breathing apparatus. More specifically, the present disclosure relates to a breathing apparatus and a method of communicating using the breathing apparatus.

BACKGROUND

(2) Personal protective equipment (PPE), such as respiratory protection devices, may be used by emergency personnel, for example, firefighters, law enforcement, first responders, healthcare professionals, paramedics or other personnel who work in potentially hazardous environments, for example, chemical, biological, or nuclear environments, or fires. While a large variety of respiratory protection devices are available, some commonly used devices may include powered air purifying respirators (PAPR) and a self-contained breathing apparatus (SCBA).

(3) Respiratory protection devices may typically include a purge valve that may fluidly communicate a facepiece or the PPE with ambient environment. Such valves may be helpful in conditions where sufficient flow of a breathing gas is not provided by the facepiece to a user. When the purge valve is actuated by the user, air may directly flow from the outside to the facepiece. This generally produces a loud hissing noise inside the facepiece known as purge noise.

(4) The personnel using the respiratory protection devices may often rely on an onboard voice communication system as it becomes difficult to conduct face-to-face communication or wireless communication of speech when the facepiece or the PPE is worn by the user. The voice communication system may provide, for example, voice amplification, transmission and/or radio communication functionality. However, any communication system may be severely degraded by background noise.

(5) Conventional voice communication systems generally detect the purge noise inside the facepiece and subsequently disable voice communication through the facepiece to mitigate any purge noise from being transmitted. However, this is not desirable as the user may need to communicate when the purge valve is actuated by the user.

SUMMARY

(6) In one aspect, a method of communicating using a breathing apparatus is described. The method includes receiving an audio signal from a sound acquisition unit. The method further includes determining a state of the breathing apparatus based on the received audio signal. The state is at least one of a first state and a second state. The method further includes applying a first filter on the audio signal if the determined state is the first state. The first filter has a first frequency response. The method further includes applying a second filter on the audio signal if the determined state is the second state. The second filter has a second frequency response different from the first frequency response of the first filter. The method further includes generating an output signal based on the application of the first filter or the second filter on the audio signal. The method further includes receiving the output signal at an output device.

(7) In another aspect, a breathing apparatus is described. The breathing apparatus includes a facepiece, an audio processing unit, and an output device. The sound acquisition unit is configured to generate an audio signal in response to a sound inside the facepiece. The audio processing unit is configured to receive the audio signal from the sound acquisition unit. The audio processing unit is further configured to determine a state of the breathing apparatus based on the received audio signal. The state is at least one of a first state and a second state. The audio processing unit is

further configured to apply a first filter on the audio signal if the determined state is the first state. The first filter has a first frequency response. The audio processing unit is further configured to apply a second filter on the audio signal if the determined state is the second state. The second filter has a second frequency response different from the first frequency response of the first filter. The audio processing unit is further configured to generate an output signal based on the application of the first filter or the second filter on the audio signal. The output device receives the output signal from the audio processing unit.

(8) In a further aspect, a method of communicating using a breathing apparatus having a facepiece and a purge valve is described. The method includes receiving an audio signal from a sound acquisition unit. The method further includes determining a state of the breathing apparatus based on the received audio signal. The state is at least one of a purge-off state and a purge-on state. In the purge-off-state, the purge valve is closed. Further, in the purge-on state, the purge valve is at least partially open. The purge valve fluidly communicates the facepiece with an ambient when the purge valve is at least partially open. The method further includes applying a first filter on the audio signal if the determined state is the purge-off state. The first filter has a first frequency response. The method further includes applying a second filter on the audio signal if the determined state is the purge-on state. The second filter has a second frequency response different from the first frequency response of the first filter. The method further includes generating an output signal based on the application of the first filter or the second filter on the audio signal. The method further includes receiving the output signal at an output device.

(9) The details of one or more examples of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Exemplary embodiments disclosed herein may be more completely understood in consideration of the following detailed description in connection with the following figures. The figures are not necessarily drawn to scale. Like numbers used in the figures refer to like components. However, it will be understood that the use of a number to refer to a component in a given figure is not intended to limit the component in another figure labeled with the same number.

(2) FIG. 1 illustrates a schematic view of an example of a breathing apparatus, in accordance with techniques of this disclosure.

(3) FIGS. 2A-2B illustrate schematic views of an example of a regulator of the breathing apparatus of FIG. 1, in accordance with techniques of this disclosure.

(4) FIG. 3 is a block diagram illustrating an exemplary breathing apparatus, in accordance with techniques of this disclosure.

(5) FIG. 4 illustrates an example of a history window of an audio signal, in accordance with techniques of this disclosure.

(6) FIG. 5 illustrates an example of an audio signal, in accordance with techniques of this disclosure.

(7) FIGS. 6A-6D illustrate various examples of a frequency response of a first filter, in accordance with techniques of this disclosure.

(8) FIG. 7 illustrates an example of a frequency response of a second filter, in accordance with techniques of this disclosure.

(9) FIG. 8 is a flow chart illustrating a method of communicating using a breathing apparatus, in accordance with techniques of this disclosure.

(10) FIG. 9 is a flow chart illustrating a method of communicating using a breathing apparatus, in

accordance with techniques of this disclosure.

(11) FIG. 10 is a flow chart illustrating a method of communicating using a breathing apparatus, in accordance with techniques of this disclosure.

DETAILED DESCRIPTION

(12) In the following description, reference is made to the accompanying figures that form a part thereof and in which various embodiments are shown by way of illustration. It is to be understood that other embodiments are contemplated and may be made without departing from the scope or spirit of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense.

(13) The term “audio signal” as used herein refers to a representation of sound in electrical form. An audio signal may be represented in the form of electrical voltage for analog signals, or a series of binary numbers for digital signals.

(14) The term “microphone” as used herein refers to a transducer or sensor that converts sound into an electrical audio signal.

(15) Unless otherwise mentioned, the term “filter” as used herein refers to an audio filter. An audio filter is a frequency dependent amplifier circuit that operates in audio frequency range. The audio filter may attenuate or amplify a particular frequency range based on the characteristics of the audio filter. The term “filter” also refers to an audio filter that provides a single filter function, such as a low-pass, high-pass, band-pass, or band-reject filter function.

(16) The term “low pass filter” as used herein denotes any device that passes signals in a low-frequency band but blocks signals in a high-frequency band. The bands may be defined based on a cutoff frequency. Conversely, the term “high pass filter” denotes any device that passes signals in the high-frequency band but blocks signals in the low-frequency band.

(17) The term “equalization filter” as used herein refers to any device that adjusts a balance between the frequency components within an audio signal.

(18) As used herein, the term “frequency response” of a system (e.g., a transducer) or an audio filter refers to an output-to-input ratio of the system or the audio filter as a function of frequency. The frequency response is typically characterized by the magnitude of the system's (or the audio filter's) transfer function, measured in dB, versus frequency, measured in Hz. Further, the term “frequency response” may describe a bandwidth, a phase response, and/or a sensitivity of the system or the audio filter over a particular bandwidth.

(19) The term “history window” as used herein refers to a period of time for which an audio signal is under consideration.

(20) The term “time frame” as used herein refers to a time period within the history window.

(21) According to aspects of this disclosure, a method of communicating using a breathing apparatus includes receiving an audio signal from a sound acquisition unit. The method further includes determining a state of the breathing apparatus based on the received audio signal. The state is at least one of a first state and a second state. The method further includes applying a first filter on the audio signal if the determined state is the first state. The first filter has a first frequency response. The method further includes applying a second filter on the audio signal if the determined state is the second state. The second filter has a second frequency response different from the first frequency response of the first filter. The method further includes generating an output signal based on the application of the first filter or the second filter on the audio signal. The method further includes receiving the output signal at an output device.

(22) The first state may refer to a state of the breathing apparatus where the purge valve is closed, while the second state may refer to a state of the breathing apparatus where the purge valve is at least partially open. Purge noise may be generated when the purge valve is at least partially open. The first filter may mitigate noise and allow speech transmission when the purge valve is closed. Further, the second filter may cut off a higher frequency range and may allow a lower frequency range to pass through. The purge noise, which typically falls in the high frequency range, may then

be suppressed while allowing speech transmission (which falls in the lower frequency range) to pass through the second filter. This may restrict purge noise from being transmitted in the output signal while allowing speech of a user to pass through even when the purge valve is at least partially open.

(23) FIG. 1 illustrates a schematic view of an example of a breathing apparatus **100**. The breathing apparatus **100** is intended to be worn by a user. In some examples, the breathing apparatus **100** may be used by emergency personnel, e.g., firefighters, law enforcement, medical personnel, first responders, paramedics, or other personnel who work in potentially hazardous environments, e.g., chemical, biological or nuclear environments, fires, or other physical environments, e.g., construction sites, mining or manufacturing sites.

(24) In some examples, the breathing apparatus **100** may be a part of a personal protective equipment (PPE). Examples of PPE may include, but are not limited to, respiratory protection equipment (including disposable respirators, reusable respirators, powered air purifying respirators, and supplied air respirators), protective eyewear, such as visors, goggles, filters or shields (any of which may include augmented reality functionality), protective headwear, such as hard hats, hoods or helmets, hearing protection (including ear plugs and ear muffs), protective shoes, protective gloves, other protective clothing, such as coveralls and aprons, protective articles, such as sensors, safety tools, detectors, global positioning devices, mining cap lamps, fall protection harnesses, exoskeletons, self-retracting lifelines, heating and cooling systems, gas detectors, and any other suitable gear configured to protect the user from injury.

(25) The breathing apparatus **100** includes a facepiece **102** having a face blank **104** and a rear opening **105** which seals around a face of a user. The face blank **104** may further include a chin portion **106** that seals around a chin area of the user. Further, the face blank **104** may include side sections that seal around respective sides of the face of the user and a forehead portion **108**, opposite the chin portion **106**, which seals around a forehead of the user. In some examples, the face blank **104** may be fabricated, for instance, from a flexible material, such as rubber, silicon, foam, plastic, and the like. In some examples, the face blank **104** may be formed of a material that is selected to be substantially impermeable to airborne environmental hazards that the breathing apparatus **100** may be designed to offer a barrier to.

(26) In some examples, the face blank **104** may be designed to form a seal at its periphery with the face of the user. Further, the facepiece **102** may cover substantially an entire face of the user. The face blank **104** further includes a series of cooperative straps **110** that are affixed to the facepiece **102** to provide a means by which the user is able to forcibly bring the facepiece **102** into contact with the face of the user to effect a seal therewith. In some examples, the straps **110** may be elasticized to ensure a continuing seal, notwithstanding movement of the user. The straps **110** includes clamps **112** that may allow the breathing apparatus **100** to be adjusted for loose fit on the face of the user.

(27) The face blank **104** may include a lens opening at an upper portion of the facepiece **102**. A transparent medium **114** is disposed in the lens opening. The transparent medium **114** may be coupled to the face blank **104** using one or more methods. For example, the transparent medium **114** and the face blank **104** may be assembled using at least one of an adhesive, an interference fit, a fastener (e.g., clip, latch, and the like), or any other fastening means. In some examples, the transparent medium **114** may be configured to provide a substantially full field of view to the user. In some examples, the transparent medium **114** may be a substantially unitary piece of material (e.g., polycarbonate) having a curved contour. For example, the transparent medium **114** may be molded from a single type of material and then coated (e.g., silicone) to protect the transparent medium **114**. The transparent medium **114** may have an interior side that faces the user and defines an interior space **116** of the facepiece **102** between the transparent medium **114** and the face of the user.

(28) The breathing apparatus **100** further includes a regulator **118**. In the illustrated example of

FIG. 1, the regulator **118** is mounted at a lower portion of the facepiece **102**. The regulator **118** may deliver a breathing gas to the facepiece **102** from a supply line **122** of pressurized gas. In some examples, the regulator **118** may reduce the incoming breathing gas to a designated pressure that is suitable for breathing by the user. In some examples, the breathing gas may be delivered via the supply line **122** at an intermediate pressure from a high-pressure source, such as a tank or a cylinder of gas (not shown). In some examples, the breathing apparatus **100** may be a self-contained breathing apparatus (SCBA) or a powered air-purifying respirator (PAPR) wherein pressurized gas may be supplied through the supply line **122** from the high-pressure source.

(29) The regulator **118** is configured to be in fluid communication with an adaptor **120**. In some examples, the regulator **118** may be removably coupled to the adaptor **120**. The adaptor **120** may be attached to the facepiece **102**. The pressurized breathing gas from the high-pressure source may be delivered to an inlet opening **123** on the regulator **118**. A further passage (not shown) may conduct the breathing gas from the regulator **118** through the adaptor **120** and deliver the breathing gas into the interior space **116** of the facepiece **102**. In some examples, the pressurized gas may be conveniently provided to the interior space **116**, such that condensation and other moisture, including exhalation moisture, may be diminished and a defogging of the transparent medium **114** may be effected.

(30) A nose cup **124** is disposed in the interior space **116**. The nose cup **124** may be configured to cover the nose and mouth of the user in a sealing manner. In the illustrated example of FIG. 1, the nose cup **124** serves as a fluid divider which separates an outside portion from an inside portion of the nose cup **124**. As used herein, the term “outside portion” may refer to the interior space **116** of the facepiece **102** between the transparent medium **114** and the face of the user. The term “inside portion” may refer to a space between the nose cup **124** and the face of the user.

(31) The nose cup **124** includes one or more inhalation valves **126**. In some examples, the inhalation valve **126** may be configured as a check valve of disk type that enables flow from the outside to the inside portion of the nose cup **124** while preventing flow in the opposite direction. In some examples, the inhalation valve **126** may have other configurations, such as a poppet, a mushroom or a flapper valve. A negative pressure may be generated inside the interior space **116** of the facepiece **102** due to inhalation of breathing gas by the user. The negative pressure generated within the interior space **116** may draw the breathing gas from the regulator **118** to the interior space **116** responsive to the breathing of the user.

(32) In some examples, the nose cup **124** may include a central opening (not shown) which is open through the lower portion of the facepiece **102** to a chamber (not shown) in the adaptor **120**. The chamber of the adaptor **120** may include an outlet **128**. Alternatively, the outlet **128** may be disposed on the regulator **118**. Airflow through the outlet **128** may be controlled by an exhalation valve (not shown). The exhalation valve may be configured to allow air to escape from the chamber of the adaptor **120** when the pressure in the chamber is at a predetermined level above the ambient atmospheric pressure. Air exhaled by the user may pass through the central opening of the nose cup **124** into the chamber of the adaptor **120**. Air exhaled by the user may then flow out of the adaptor **120** through the exhalation valve and the outlet **128**.

(33) In some examples, the exhalation valve may be configured as a disk type valve that is spring loaded to a closed position. In other examples, the exhalation valve may have other configurations such as a poppet, a mushroom or a flapper valve. In some examples, the exhalation valve may be configured to prevent airflow from the atmosphere to enter the chamber inside the adaptor **120**.

(34) In some examples, the breathing apparatus **100** may further include accessories such as voice emitters, filtering components, etc. For example, the voice emitters may be mounted on the breathing apparatus **100** to amplify the voice of the user to facilitate communication with other individuals and help provide intelligible speech transmittance through the breathing apparatus **100**. In some examples, the breathing apparatus **100** may further include a voice communication system that provides, for example, voice amplification, transmission and/or radio communication

functionality.

(35) FIGS. 2A-2B illustrate a schematic view of an example of a regulator **218** of a breathing apparatus **100**. The regulator **218** may be similar to the regulator **118** of the breathing apparatus **100** of FIG. 1. The regulator **218** may be removably coupled to an adaptor **220**. The adaptor **220** may be similar to the adaptor **120** of FIG. 1. Referring now to FIGS. 1 and 2A, the adaptor **220** includes a spray bar **222** having a plurality of gas delivery openings (not shown) through which the breathing gas may enter the interior space **116** of the facepiece **102**. In some examples, the gas delivery openings of the spray bar **222** may be positioned in the interior space **116** of the facepiece **102** outside the nose cup **124**.

(36) The regulator **218** further includes a purge valve **230** that fluidly communicates the facepiece **102** with an ambient when the purge valve **230** is at least partially open. In some examples, the purge valve **230** may allow air to enter through an opening **232** of the purge valve **230** and flow into the facepiece **102** when the purge valve **230** is at least partially open. The purge valve **230** may bypass the flow of breathing gas from the high-pressure source through an inlet opening **223** on the regulator **218**. In some examples, the inlet opening **223** may be similar to the inlet opening **123** of FIG. 1.

(37) In some examples, the purge valve **230** may include an actuator **234** coupled to a valve member **236**. In the example shown in FIG. 2A, the valve member **236** covers the opening **232** of the purge valve **230** restricting air from entering through the opening **232**. This may represent a first state **240** of the breathing apparatus **100**. In some examples, the first state **240** may correspond to a purge-off state of the breathing apparatus **100**. The actuator **234** may be actuated to control the valve member **236** and selectively open or close the opening **232**. In some examples, the actuator **234** may be configured as a manually controlled knob. For example, the opening **232** of the purge valve **230** may be opened by manually rotating a control portion of a knob. In some examples, the actuator **234** of the purge valve **230** may be configured as a purge button. In some examples, the purge valve **230** may be actuated manually. In other examples, the purge valve **230** may be actuated automatically. It should be understood that any form or configuration of the purge valve **230**, the actuator **234** and the valve member **236** may be utilized based on application requirements without limiting the scope of the present disclosure.

(38) FIG. 2B illustrates a schematic view of the regulator **218** in which the purge valve **230** is at least partially open. This may represent a second state **242** of the breathing apparatus **100**. In some examples, the second state **242** may correspond to a purge-on state of the breathing apparatus **100**. In the illustrated example, the valve member **236** may be controlled by the actuator **234** to allow air to flow through the opening **232** of the purge valve **230** in the open state of the purge valve **230**. Referring to FIGS. 1 and 2B, air may flow towards the spray bar **222** and into the interior space **116** of the facepiece **102** through the plurality of gas delivery openings (not shown) of the spray bar **222**. In some examples, the opening **232** may be partially or fully covered by the valve member **236**. In some examples, as the air flows towards the facepiece **102**, it may produce a noise inside the facepiece **102**. The noise may be referred to as a purge noise.

(39) In some examples, the purge valve **230** may be configured as a safety valve which may be actuated by the user during emergency conditions. For example, the user may actuate the purge valve **230** when sufficient flow of breathing gas is not provided by the regulator **218**. When the purge valve **230** is actuated through the actuator **234**, air may freely flow to the interior space **116** of the facepiece **102** and may be inhaled by the user through the one or more inhalation valves **126**. Additionally, or alternatively, the purge valve **230** may also be actuated to mitigate fogging of the transparent medium **114** from inside the facepiece **102** due to condensation of moisture, for example, due to exhalation by the user.

(40) FIG. 3 is a block diagram illustrating an exemplary breathing apparatus **300**. The breathing apparatus **300** may be similar to the breathing apparatus **100** of FIGS. 1, 2A and 2B, and similar reference numbers are used to designate same or similar elements. Referring to FIGS. 1, 2A, 2B

and 3, the breathing apparatus **300** includes the facepiece **102** having a sound acquisition unit **302** configured to generate an audio signal A in response to a sound inside the facepiece **102**. The sound acquisition unit **302** may be suitably configured and positioned to detect the sound inside the facepiece **102**, for example, a sound of user speech. The sound acquisition unit **302** may also detect other sounds associated within the breathing apparatus **300**, for example, breathing sound of the user, background noise, low pressure alarm noise, personal alert safety system noise, etc.

(41) In some examples, the sound acquisition unit **302** may include a microphone. In some examples, the sound acquisition unit **302** may include multiple microphones. As used herein, the term “microphone” refers to a transducer or sensor that converts sound into an electrical audio signal. The sound acquisition unit **302** may receive mechanical vibrations from the user speech and may convert the mechanical vibrations into electric audio signals. In some examples, the electric audio signals may comprise a speech signal corresponding to user speech, and a noise signal indicative of other sounds inside the facepiece **102**.

(42) The sound acquisition unit **302** may be mounted inside the facepiece **102** and the nose cup **124**. Alternatively, the sound acquisition unit **302** may be mounted outside the facepiece **102**. In such an arrangement, the sound acquisition unit **302** may include a diaphragm coupled to the microphone. The diaphragm may transmit mechanical vibrations from the user speech inside the facepiece **102** to the microphone outside the facepiece **102**. It should be understood that other arrangements of the sound acquisition unit **302** are also possible and are within the scope of the present disclosure. A user, for example a firefighter, typically wears the breathing apparatus **300** in an emergency situation, and therefore his or her face may be tightly covered by the facepiece **102**. When the user starts to speak, the voice may generate positive pressure inside the facepiece **102** that may be picked up by the sound acquisition unit **302**.

(43) The breathing apparatus **300** further includes an audio processing unit **304** configured to receive the audio signal A from the sound acquisition unit **302**. The audio signal A may also be interchangeably referred to as “the received audio signal A”. In some examples, the audio processing unit **304** may process the detected sound and deliver a processed speech to an amplifier and/or a speaker for face-to-face communication and to a wireless communication interface for wireless communications. For example, the audio processing unit **304** may eliminate breathing sound of the user from being transmitted outside the facepiece **102**. Further, the breathing apparatus **300** may be equipped with noise cancellation functionality that may cancel an ambient noise.

(44) The audio processing unit **304** is configured to determine a state S of the breathing apparatus **300** based on the received audio signal A. The state S may also be interchangeably referred to as “the determined state S”. The state S is at least one of the first state **240** and the second state **242** (refer to FIGS. 2A and 2B). In some examples, the first state **240** may correspond to a purge-off state when the purge valve **230** is closed and the second state **242** may correspond to a purge-on state when the purge valve **230** is at least partially open. As described previously, the purge valve **230** may be configured to be actuated by the user of the breathing apparatus **300**. The purge valve **230** may be configured to be partially or fully opened by the user. Thus, the purge valve **230** may have different configurations, for example, closed, partially open, and completely open. Further, the purge valve **230** may fluidly communicate the facepiece **102** with the ambient when the purge valve **230** is at least partially open and air may flow from the outside to the interior space **116** of the facepiece **102**. This flow of air may produce a noise inside the facepiece **102**, for example, a loud hissing noise.

(45) The state S of the breathing apparatus **300** may be determined by the audio processing unit **304** and will be described in detail hereinafter. Specifically, the state S of the breathing apparatus **300** may be either the purge-on state or the purge-off state. In some examples, the audio processing unit **304** may be configured to obtain the audio signal A over a history window. The audio signal A may be generated by the sound acquisition unit **302**. FIG. 4 illustrates an example of a history window **400** of an audio signal A. The history window **400** may correspond to a first time period

T1. As used herein, “history window” refers to a period of time for which an audio signal **A** is under consideration. For example, the history window **400** or the first time period **T1** may correspond to a period of 30 milliseconds.

(46) In some examples, the audio processing unit **304** may include a processor and suitable circuitry for carrying out various functions of the audio processing unit **304**. The circuitry may include one or more analog-to-digital converters, a digital-to-analog converters, input/output ports, buses, and so forth. In some cases, the audio processing unit **304** may be embodied as a printed circuit board including various electronic components. In some examples, the audio processing unit **304** may further include a memory to store the history window **400**. In some examples, the memory may be a main memory, a static memory, or a dynamic memory. In some examples, the memory may include, but may not be limited to, a computer readable storage media, such as various types of volatile and non-volatile storage media, including but not limited to, random access memory, read-only memory, programmable read-only memory, electrically programmable read-only memory, electrically erasable read-only memory, flash memory, and the like.

(47) The audio processing unit **304** may be further configured to divide the history window **400** into a plurality of time frames **402**. As used herein, the term “time frame” may refer to a time period within the history window **400**. Each time frame **402** may correspond to a second time period **T2** less than the first time period **T1** (i.e., $T2 < T1$). In some examples, the second time period **T2** may correspond to a period of 3 milliseconds. The second time period **T2** may be determined depending on factors, such as a cycle of the breath of the user. For example, the history window **400** may comprise 10 of the time frames **402** with the first time period **T1** of 30 milliseconds and the second time period **T2** of 3 milliseconds. The time frames **402** shown in FIG. 4 are exemplary in nature. It should be understood that the values of the first time period **T1** and the second time period **T2** are by way of example only, and the values may change as per the application requirements.

(48) The audio processing unit **304** may be further configured to compare an amplitude **AM** of the audio signal **A** in the plurality of time frames **402** with a predetermined noise threshold **404**. In other words, the amplitude **AM** of the audio signal **A** in each time frame **402** may be compared with the predetermined noise threshold **404**. Specifically, the amplitude **AM** of the audio signal **A** may be above or below the predetermined noise threshold **404**. The predetermined noise threshold **404** may correspond to the noise generated due to actuation of the purge valve **230**, for example, the purge noise. The predetermined noise threshold **404** may be selected based on an amplitude of the purge noise. In some examples, the predetermined noise threshold **404** may be subjected to modifications as per application requirements. In some cases, the predetermined noise threshold **404** may be selected based on operating conditions of the breathing apparatus **300**.

(49) The audio processing unit **304** may be further configured to determine a purge noise frame count **N** in the history window **400** corresponding to a number of time frames **402** in which the amplitude **AM** of the audio signal **A** exceeds the predetermined noise threshold **404**. The time frames **402** may be determined wherein the amplitude **AM** of the audio signal **A** is greater than the predetermined noise threshold **404**. Such time frames **402** may be added to the purge noise frame count **N**.

(50) The audio processing unit **304** may be further configured to determine that the breathing apparatus **300** is in the purge-on state if the purge noise frame count **N** is greater than a predetermined purge count threshold **N1**. In some examples, the purge count threshold **N1** may be 90%. It means that if 90% of the time frames **402** in the history window **400** are included in purge noise frame count **N**, then the breathing apparatus **300** is determined to be in purge-on state. In some examples, the purge count threshold **N1** may be 80%. In other examples, the purge count threshold **N1** may be between 40% to 100%. It should be understood that the value of purge count threshold **N1** is by way of example only, and the value may change as per the application requirements. Similarly, the audio processing unit **304** may be further configured to determine that

the breathing apparatus **300** is in the purge-off state if the purge noise frame count **N** is less than or equal to the predetermined purge count threshold **N1**.

(51) FIG. 5 illustrates an example of an audio signal **410** and corresponding time frames **412**. Referring to FIGS. 3 and 5, the audio processing unit **304** may be configured to compare an amplitude **AM1** of the audio signal **410** in time frame **412** with a predetermined noise threshold (not shown). The audio processing unit **304** may determine a purge noise frame count **N** corresponding to a number of the time frames **412** in which the amplitude **AM1** of the audio signal **410** exceeds the predetermined noise threshold. The breathing apparatus **300** may be in the purge-on state if the purge noise frame count **N** is greater than the predetermined purge count threshold **N1**, for example, 90%. In the example of FIG. 5, only two time frames **412** are shown for illustration purpose only. A first frame **414** corresponds to beginning of purge-on state of the breathing apparatus **300** while a second frame **416** corresponds to purge-off state. Further, a count of the time frames **412** (not shown) between the first frame **414** and the second frame **416** may be above the predetermined purge count threshold **N1**.

(52) Referring to FIGS. 3, the audio processing unit **304** includes a first filter **306** that may process the audio signal **A** received from the sound acquisition unit **302**. As used herein, the term “filter” refers to an audio filter. An audio filter is a frequency dependent amplifier circuit that operates in audio frequency range where the audio filter may attenuate or amplify a particular frequency range based on the characteristics of the audio filter. The audio processing unit **304** is further configured to apply the first filter **306** on the audio signal **A** if the determined state **S** is the first state **240**. In some examples, the first state **240** may correspond to the purge-off state.

(53) The first filter **306** has a first frequency response **F1**. FIGS. 6A-6D illustrate various examples of the frequency response of the first filter **306**. Specifically, FIGS. 6A-6D illustrate frequency responses of first filters **306A**, **306B**, **306C** and **306D**, respectively. The first filters **306A-306D** may be collectively referred to as the first filter **306**. In some examples, the first frequency response **F1** of the first filter **306** is inverse of a frequency response **F** (shown in FIG. 6A) of the facepiece **102** (shown in FIG. 1). In general, the facepiece **102** may have a characteristic frequency response and may amplify or attenuate certain frequencies corresponding to its characteristic frequency response. The frequency response **F** corresponds to the characteristic frequency response of the facepiece **102**. The first filter **306** may modify the audio signal **A** opposite to the frequency response **F** of the facepiece **102**. Thus, the first filter **306** may amplify a given frequency range for which the facepiece **102** may attenuate the audio signal **A**, and/or vice versa.

(54) In the illustrated example of FIG. 6A, the first filter **306A** may amplify frequencies within a boost frequency range. The boost frequency range may be greater than about 1.5 kilohertz (kHz) and may be less than about 3 kHz. The first filter **306A** may further attenuate frequencies less than about 1 kHz. In some examples, the first filter **306A** may reduce the noise generated from the breathing of the user (that may typically be in the range of about 1 kHz) from being transmitted from the facepiece **102** since the first filter **306A** may suppress frequencies less than about 1 kHz. This may reduce speech distortion and render realistic and clearer audio to be transmitted from the facepiece **102**. In some examples, the first filter **306A** may be an equalization filter. The equalization filter may adjust a balance between the frequency components within the audio signal **A**. After equalization, the user speech intelligibility may be improved.

(55) Each of the first filters **306A**, **306B**, **306C**, **306D** also amplify frequencies at least in a range from about 1.5 kHz to about 3 kHz. However, in the illustrated example of FIG. 6B, the first filter **306B** does not attenuate or amplify frequencies less than about 1 kHz. In the illustrated example of FIG. 6C, the first filter **306C** amplifies frequencies below 1 kHz. In the illustrated example of FIG. 6D, the first filter **306D** amplifies frequencies below 1 kHz to a greater extent than the first filter **306C** of FIG. 6C.

(56) The audio processing unit **304** may be further configured to select the first filter **306** from a set of first audio filters. In some examples, the set of first audio filters may comprise audio filters with

different frequency responses, such as the examples shown in FIGS. 6A-6D. Specifically, the audio processing unit **304** may be further configured to select the first filter **306** from the set of first filters **306A**, **306B**, **306C**, **306D**. For example, some audio filters may amplify a particular frequency range and others may attenuate a particular frequency range. In some examples, the one or more first filters **306** from the set of first audio filters may modify bass, treble and/or middle sections of the audio signal A. In some examples, more than one first filter **306** from the set of first audio filters may be selected. In some cases, the audio processing unit **304** may select one of the first filters **306A**, **306B**, **306C**, **306D** as the first filter **306** based on a voice parameter (e.g., pitch) of a user.

(57) Referring again to FIG. 3, the audio processing unit **304** may further include a second filter **308** that may process the audio signal A received from the sound acquisition unit **302**. The audio processing unit **304** is further configured to apply the second filter **308** on the audio signal A if the determined state S is the second state **242**. In some examples, the second state **242** may correspond to the purge-on state. The second filter **308** has a second frequency response F2 different from the first frequency response F1 of the first filter **306**. FIG. 7 illustrates an example of the second frequency response F2 of the second filter **308**. In the illustrated example, the second filter **308** may be a low pass filter. In the illustrated example, the second filter **308** may have a cut-off frequency CF of about 3 kHz. Thus, the second filter **308** may allow a low frequency portion of the audio signals A with a frequency less than 3 kHz to pass through the second filter **308** and may attenuate a high frequency portion of the audio signal A with frequencies greater than 3 kHz. Hence, the second filter **308** may suppress frequencies greater than 3 kHz, for example, the purge noise. The second filter **308** may therefore prevent the purge noise from being transmitted from the breathing apparatus **300**. The second filter **308** may allow speech of a user to be transmitted during the purge-on state of the breathing apparatus **300** since the second filter **308** allows lower frequencies to pass through. However, in conventional systems, the voice transmission is suppressed entirely during purge-on state of a breathing apparatus.

(58) In some examples, the audio processing unit **304** may be further configured to select the second filter **308** from a set of second audio filters. In some examples, the set of second audio filters may comprise audio filters with different frequency responses. For example, some audio filters may amplify a particular frequency range and others may attenuate a particular frequency range. In some examples, the one or more second filters **308** from the set of second audio filters may modify bass, treble and/or middle sections of the audio signal A. In some examples, more than one second filter **308** from the set of second audio filters may be selected.

(59) The audio processing unit **304** may be further configured to clear the history window **400** (shown in FIG. 4). The audio processing unit **304** may repeat the process of determining the state S of the breathing apparatus **300** by obtaining a new history window **400**. The audio processing unit **304** may periodically clean the history window **400** from the memory.

(60) Referring again to FIG. 3, the audio processing unit **304** is further configured to generate an output signal O based on the application of the first filter **306** or the second filter **308** on the audio signal A. The breathing apparatus **300** further includes an output device **310** receiving the output signal O from the audio processing unit **304**. In some examples, the output device **310** may include a speaker. In some examples, the output device **310** may include multiple speakers. The audio signal A may, for example, be transmitted to an amplifier and/or a speaker to facilitate face-to-face communication by the user with others in the vicinity of the user.

(61) In some examples, the breathing apparatus **300** may further include a wireless unit (not shown) configured to wirelessly transmit the output signal O from the audio processing unit **304** to the output device **310**. For instance, the audio signal A may be transmitted for communication by the user to others remote from the user. Such a wireless unit may utilize a wireless communication interface, for example, Bluetooth®, or any other wireless network. As used herein, the phrase “wireless communication interface” refers to a system interface or a network interface between the

breathing apparatus **300** and the output device **310** in a network, for example, a wireless radio network. Alternatively, the output signal O may be transmitted from the audio processing unit **304** to the output device **310** using a wired communication system.

(62) In some examples, the wireless unit may be used to transmit the output signal O to a portable handheld device used for wireless communication between users. In such cases, the output device **310** may be part of the portable handheld device. For example, the wireless unit may transmit the output signal O to a two-way handheld radio transceiver, such as a walkie-talkie.

(63) FIG. **8** is a flow chart illustrating a method **500** of communicating using a breathing apparatus. The method **500** may be implemented using the breathing apparatus **300** described above. The breathing apparatus **300** may be similar to the breathing apparatus **100** of FIGS. **1** and **2A-2B** and similar reference numbers are used to designate same or similar elements. In some examples, the breathing apparatus **300** may be a self-contained breathing apparatus (SCBA) or a powered air-purifying respirator (PAPR). Referring to FIGS. **1-8**, at block **502**, the method **500** includes receiving the audio signal A from the sound acquisition unit **302**. In some examples, the sound acquisition unit **302** may include the microphone. In some examples, the sound acquisition unit **302** may further include the diaphragm coupled to the microphone.

(64) At block **504**, the method **500** further includes determining the state S of the breathing apparatus **300** based on the received audio signal A. The state S is at least one of the first state **240** and the second state **242**. In some examples, the first state **240** may correspond to the purge-off state and the second state **242** may correspond to the purge-on state of the breathing apparatus **300**. Further, in the purge-off state, the purge valve **230** of the breathing apparatus **300** may be closed. In the purge-on state, the purge valve **230** of the breathing apparatus **300** may be at least partially open. In some examples, the purge valve **230** may fluidly communicate the facepiece **102** of the breathing apparatus **300** with the ambient when the purge valve **230** is at least partially open.

(65) In some examples, determining the state S of the breathing apparatus **300** may further include obtaining the audio signal A over the history window **400**. The history window **400** may correspond to the first time period T1. In some examples, determining the state S of the breathing apparatus **300** may further include dividing the history window **400** into the plurality of time frames **402**. Each time frame **402** may correspond to the second time period T2 less than the first time period T1. In some examples, determining the state S of the breathing apparatus **300** may further include comparing the amplitude AM of the audio signal A in the plurality of time frames **402** with the predetermined noise threshold **404**. In some examples, determining the state S of the breathing apparatus **300** may further include determining the purge noise frame count N in the history window **400** corresponding to the number of time frames **402** in which the amplitude AM of the audio signal A exceeds the predetermined noise threshold **404**.

(66) In some examples, the method **500** may further include determining that the breathing apparatus **300** is in the purge-on state if the purge noise frame count N is greater than the predetermined purge count threshold N1. In some examples, the method **500** may further include determining that the breathing apparatus **300** is in the purge-off state if the purge noise frame count N is less than or equal to the predetermined purge count threshold N1.

(67) Control moves to block **506** if the determined state S is the first state **240**. Otherwise, control moves to block **508** if the determined state S is the second state **242**.

(68) At block **506**, the method **500** further includes applying the first filter **306** on the audio signal A if the determined state S is the first state **240**. In some examples, the first state **240** may correspond to the purge-off state. The first filter **306** has the first frequency response F1. The method **500** may further include selecting the first filter **306** from the set of first audio filters, for example, the first filters **306A**, **306B**, **306C**, **306D**. In some examples, the first frequency response F1 of the first filter **306** may be inverse of the frequency response F of the facepiece **102** of the breathing apparatus **300**. In some examples, the first filter **306** may amplify frequencies within the boost frequency range. In some examples, the boost frequency range may be greater than about 1.5

kHz and less than about 3 kHz. In some examples, the first filter **306** may further attenuate frequencies less than about 1 kHz. In some other examples, the first filter **306** may further amplify frequencies less than about 1 kHz.

(69) At block **508**, the method **500** further includes applying the second filter **308** on the audio signal A if the determined state S is the second state **242**. In some examples, the second state **242** may correspond to the purge-on state. The second filter **308** has the second frequency response F2 different from the first frequency response F1 of the first filter **306**. In some examples, the method **500** may further include selecting the second filter **308** from the set of second audio filters. In some examples, the second filter **308** may be the low pass filter. In some examples, the second filter **308** may have the cut-off frequency CF of about 3 kHz.

(70) Control moves to block **510** from either of blocks **506** or **508**.

(71) At block **510**, the method **500** further includes generating the output signal O based on the application of the first filter **306** or the second filter **308** on the audio signal A. At block **512**, the method **500** further includes receiving the output signal O at the output device **310**. In some examples, the output device **310** may include the speaker. In some examples, the method **500** may further include wirelessly transmitting the output signal O to the output device **310**.

(72) The second state **242** may refer to purge-on state of the breathing apparatus **300** while the first state **240** may refer to purge-off state of the breathing apparatus **300**. The second filter **308** may advantageously cut off a higher frequency range and may allow a lower frequency range to pass through. The purge noise, which typically falls in the higher frequency range, may then be suppressed while allowing speech transmission (which typically falls in the lower frequency range) to pass through the second filter **308**. This may restrict purge noise from being transmitted in the output signal O to the output device **310** while allowing speech of a user to pass through.

(73) In some examples, the method **500** may further include clearing the history window **400**. The audio processing unit **304** may repeat the process by receiving the audio signal A again at block **502** and generating the output signal O to be received at the output device **310**.

(74) FIG. **9** is a flow chart illustrating a method **600** of communicating using a breathing apparatus having a facepiece and a purge valve. The method **600** may be implemented using the breathing apparatus **300** described above. The breathing apparatus **300** may be similar to the breathing apparatus **100** of FIGS. **1** and **2A-2B** and similar reference numbers are used to designate same or similar elements. Referring to FIGS. **1-7** and **9**, the method **600** starts at block **601**. Control moves to block **602** upon start. At block **602**, the method **600** includes receiving the audio signal A from the sound acquisition unit **302**. At block **604**, the method **600** further includes determining the state S of the breathing apparatus **300**. The state S is at least one of the purge-off state and the purge-on state. In the purge-off-state, the purge valve **230** is closed. In the purge-on state, the purge valve **230** is at least partially open. A default state of the breathing apparatus **300** is purge-off state and a default filter selected by the audio processing unit **304** is the first filter **306**. The purge valve **230** fluidly communicates the facepiece **102** with the ambient when the purge valve **230** is at least partially open.

(75) At block **604**, control moves to block **606** if the breathing apparatus **300** is in the purge-off state. Otherwise, control moves to block **616** if the breathing apparatus **300** is in the purge-on state.

(76) In some examples, determining the state S of the breathing apparatus **300** may further include obtaining the audio signal A over the history window **400**. The history window **400** may correspond to the first time period T1. In some examples, determining the state S of the breathing apparatus **300** may further include dividing the history window **400** into the plurality of time frames **402**. Each time frame **402** may correspond to the second time period T2 less than the first time period T1. In some examples, determining the state S of the breathing apparatus **300** may further include comparing the amplitude AM of the audio signal A in the plurality of time frames **402** with the predetermined noise threshold **404**. In some examples, determining the state S of the breathing apparatus **300** may further include determining the purge noise frame count N, at block **606**, in the

history window **400** corresponding to the number of time frames **402** in which the amplitude AM of the audio signal A exceeds the predetermined noise threshold **404**.

(77) At block **608**, the method **600** may further include determining that the breathing apparatus **300** is in the purge-on state if the purge noise frame count N is greater than the predetermined purge count threshold N1. If the purge noise frame count N is not greater than the predetermined purge count threshold N1, control may move to block **626**. At block **626**, control may return to block **602**. In some cases, the method **600** may end at block **626** if the breathing apparatus **300** is deactivated.

(78) Control moves to block **610** if the purge noise frame count N is greater than the predetermined purge count threshold N1. At block **610**, the state S of the breathing apparatus **300** is set to purge-on state if the purge noise frame count N is greater than the predetermined purge count threshold N1. At block **612**, the method **600** further includes applying the second filter **308** on the audio signal A if the determined state S is the purge-on state. The audio processing unit **304** may switch the default filter (e.g., the first filter **306**) to the second filter **308**. The second filter **308** has the second frequency response F2 different from the first frequency response F1 of the first filter **306**. In some examples, the method **600** may further include selecting the second filter **308** from the set of second audio filters. At block **614**, the method **600** may further include clearing the history window **400**. Control moves to block **626** from block **614**. At block **626**, control may return to block **602**. In some cases, the method **600** may end at block **626** if the breathing apparatus **300** is deactivated.

(79) The method **600** may repeat the process by receiving the audio signal A from the sound acquisition unit **302** at block **602**. In this case, the state S of the breathing apparatus **300** is now set to purge-on state. Hence, control may move to block **616** from block **604**. The audio processing unit **304** may further determine the state S of the breathing apparatus **300** by obtaining the audio signal A over a new history window **400**. The history window **400** may again be divided into the plurality of time frames **402**. Further, the amplitude AM of the audio signal A in the plurality of time frames **402** may be compared with the predetermined noise threshold **404**. The purge noise frame count N in the history window **400** may be determined, at block **616**, corresponding to the number of time frames **402** in which the amplitude AM of the audio signal A exceeds the predetermined noise threshold **404**.

(80) At block **618**, the method **600** may further include determining that the breathing apparatus **300** is in the purge-off state if the purge noise frame count N is less than or equal to the predetermined purge count threshold N1. If the purge noise frame count N is greater than the predetermined purge count threshold N1, control may move to block **626**. At block **626**, control may return to block **602**. In some cases, the method **600** may end at block **626** if the breathing apparatus **300** is deactivated.

(81) Control moves to block **620** if the purge noise frame count N is less than or equal to the predetermined purge count threshold N1. At block **620**, the state S of the breathing apparatus **300** is set to purge-off state if the purge noise frame count N is less than or equal to the predetermined purge count threshold N1. At block **622**, the method **600** further includes applying the first filter **306** on the audio signal A if the determined state S is the purge-off state. The audio processing unit **304** may switch the previously selected second filter **308** to the first filter **306**. The first filter **306** has the first frequency response F1. In some examples, the method **600** may further include selecting the first filter **306** from the set of first audio filters, for example, the first filters **306A**, **306B**, **306C**, **306D**.

(82) At block **624**, the method **600** may further include clearing the history window **400**. Control moves to block **626** from block **624**. At block **626**, control may return to block **602**. The method **600** again repeats the process by receiving the audio signal A from the sound acquisition unit **302** at block **602**. Now, the state S of the breathing apparatus **300** is set to purge-off state. In some cases, the method **600** may end at block **626** if the breathing apparatus **300** is deactivated.

(83) The method **600** further includes generating the output signal **O** based on the application of the first filter **306** or the second filter **308** on the audio signal **A**. The method **600** further includes receiving the output signal **O** at the output device **310**.

(84) FIG. **10** illustrates a method **700** of communicating using a breathing apparatus. The method **700** may be implemented using the breathing apparatus **300** described above. The breathing apparatus **300** may be similar to the breathing apparatus **100** of FIGS. **1** and **2A-2B** and similar reference numbers are used to designate same or similar elements. In some examples, the breathing apparatus **300** may be a self-contained breathing apparatus (SCBA) or a powered air-purifying respirator (PAPR).

(85) Referring to FIGS. **1-7** and **10**, at step **702**, the method **700** includes receiving the audio signal **A** from the sound acquisition unit **302**. In some examples, the sound acquisition unit **302** may include a microphone. In some examples, the sound acquisition unit **302** may further include a diaphragm coupled to the microphone.

(86) At step **704**, method **700** further includes determining the state **S** of the breathing apparatus **300** based on the received audio signal **A**. The state **S** is at least one of the first state **240** and the second state **242**. In some examples, the first state **240** may correspond to the purge-off state and the second state **242** may correspond to the purge-on state of the breathing apparatus **300**. In some examples, in the purge-off state, the purge valve **230** of the breathing apparatus **300** may be closed, and whereas, in the purge-on state, the purge valve **230** of the breathing apparatus **300** may be at least partially open. In some examples, the purge valve **230** may fluidly communicate a facepiece **102** of the breathing apparatus **300** with an ambient when the purge valve **230** is at least partially open.

(87) In some examples, determining the state **S** of the breathing apparatus **300** may further include obtaining the audio signal **A** over the history window **400**. In some examples, the history window **400** may correspond to the first time period **T1**. In some example, determining the state **S** of the breathing apparatus **300** may further include dividing the history window **400** into the plurality of time frames **402**. In some examples, each time frame **402** may correspond to the second time period **T2** less than the first time period **T1**. In some examples, determining the state **S** of the breathing apparatus **300** may further include comparing the amplitude **AM** of the audio signal **A** in the plurality of time frames **402** with the predetermined noise threshold **404**. In some examples, determining the state **S** of the breathing apparatus **300** may further include determining the purge noise frame count **N** in the history window **400** corresponding to a number of the time frames **402** in which the amplitude **AM** of the audio signal **A** exceeds the predetermined noise threshold **404**.

(88) In some examples, the method **700** may further include determining that the breathing apparatus **300** is in the purge-on state if the purge noise frame count **N** is greater than the predetermined purge count threshold **N1**. In some examples, the method **700** may further include determining that the breathing apparatus **300** is in the purge-off state if the purge noise frame count **N** is less than or equal to the predetermined purge count threshold **N1**. In some examples, the method **700** may further include clearing the history window **400**.

(89) At step **706**, the method **700** further includes applying the first filter **306** on the audio signal **A** if the determined state **S** is the first state **240**. The first filter **306** has the first frequency response **F1**. In some examples, the first frequency response **F1** of the first filter **306** may be inverse of the frequency response **F** of the facepiece **102** of the breathing apparatus **300**. In some examples, the first filter **306** may amplify frequencies within a boost frequency range. In some examples, the boost frequency range may be greater than about 1.5 kHz and less than about 3 kHz. In some examples, the first filter **306** may further attenuate frequencies less than about 1 kHz. In some other examples, the first filter **306** may further amplify frequencies less than about 1 kHz. In some examples, the method **700** may further include selecting the first filter **306** from the set of first audio filters, for example, the first filters **306A**, **306B**, **306C**, **306D**.

(90) At step **708**, the method **700** further includes applying the second filter **308** on the audio signal

A if the determined state S is the second state **242**. The second filter **308** has the second frequency response F2 different from the first frequency response F1 of the first filter **306**. In some examples, the second filter **308** may be a low pass filter. In some examples, the second filter **308** may have the cut-off frequency CF of about 3 kHz. In some examples, the method may further include selecting the second filter **308** from a set of second audio filters.

(91) At step **710**, the method **700** further includes generating the output signal O based on the application of the first filter **306** or the second filter **308** on the audio signal A. At step **712**, the method **700** further includes receiving the output signal O at the output device **310**. In some examples, the method **700** may further include wirelessly transmitting the output signal O to the output device **310**. In some examples, the output device **310** may include a speaker.

(92) In the present detailed description of the preferred embodiments, reference is made to the accompanying drawings, which illustrate specific embodiments in which the invention may be practiced. The illustrated embodiments are not intended to be exhaustive of all embodiments according to the invention. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present invention. The following detailed description, therefore, is not to be taken in a limiting sense, and the scope of the present invention is defined by the appended claims.

(93) Unless otherwise indicated, all numbers expressing feature sizes, amounts, and physical properties used in the specification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the foregoing specification and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by those skilled in the art utilizing the teachings disclosed herein.

(94) As used in this specification and the appended claims, the singular forms “a,” “an,” and “the” encompass embodiments having plural referents, unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(95) Spatially related terms, including but not limited to, “proximate,” “distal,” “lower,” “upper,” “beneath,” “below,” “above,” and “on top,” if used herein, are utilized for ease of description to describe spatial relationships of an element(s) to another. Such spatially related terms encompass different orientations of the device in use or operation in addition to the particular orientations depicted in the figures and described herein. For example, if an object depicted in the figures is turned over or flipped over, portions previously described as below or beneath other elements would then be above or on top of those other elements.

(96) As used herein, when an element, component, or layer for example is described as forming a “coincident interface” with, or being “on,” “connected to,” “coupled with,” “stacked on” or “in contact with” another element, component, or layer, it can be directly on, directly connected to, directly coupled with, directly stacked on, in direct contact with, or intervening elements, components or layers may be on, connected, coupled or in contact with the particular element, component, or layer, for example. When an element, component, or layer for example is referred to as being “directly on,” “directly connected to,” “directly coupled with,” or “directly in contact with” another element, there are no intervening elements, components or layers for example. The techniques of this disclosure may be implemented in a wide variety of computer devices, such as servers, laptop computers, desktop computers, notebook computers, tablet computers, hand-held computers, smart phones, and the like. Any components, modules or units have been described to emphasize functional aspects and do not necessarily require realization by different hardware units. The techniques described herein may also be implemented in hardware, software, firmware, or any combination thereof. Any features described as modules, units or components may be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. In some cases, various features may be implemented as an integrated circuit device, such as an

integrated circuit chip or chipset. Additionally, although a number of distinct modules have been described throughout this description, many of which perform unique functions, all the functions of all of the modules may be combined into a single module, or even split into further additional modules. The modules described herein are only exemplary and have been described as such for better ease of understanding.

(97) If implemented in software, the techniques may be realized at least in part by a computer-readable medium comprising instructions that, when executed in a processor, performs one or more of the methods described above. The computer-readable medium may comprise a tangible computer-readable storage medium and may form part of a computer program product, which may include packaging materials. The computer-readable storage medium may comprise random access memory (RAM) such as synchronous dynamic random access memory (SDRAM), read-only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, magnetic or optical data storage media, and the like. The computer-readable storage medium may also comprise a non-volatile storage device, such as a hard-disk, magnetic tape, a compact disk (CD), digital versatile disk (DVD), Blu-ray disk, holographic data storage media, or other non-volatile storage device.

(98) The term “processor,” as used herein may refer to any of the foregoing structure or any other structure suitable for implementation of the techniques described herein. In addition, in some aspects, the functionality described herein may be provided within dedicated software modules or hardware modules configured for performing the techniques of this disclosure. Even if implemented in software, the techniques may use hardware such as a processor to execute the software, and a memory to store the software. In any such cases, the computers described herein may define a specific machine that is capable of executing the specific functions described herein. Also, the techniques could be fully implemented in one or more circuits or logic elements, which could also be considered a processor.

(99) In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored on or transmitted over, as one or more instructions or code, a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media, or communication media including any medium that facilitates transfer of a computer program from one place to another, e.g., according to a communication protocol. In this manner, computer-readable media generally may correspond to (1) tangible computer-readable storage media, which is non-transitory or (2) a communication medium such as a signal or carrier wave. Data storage media may be any available media that can be accessed by one or more computers or one or more processors to retrieve instructions, code and/or data structures for implementation of the techniques described in this disclosure. A computer program product may include a computer-readable medium.

(100) By way of example, and not limitation, such computer-readable storage media can comprise RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage, or other magnetic storage devices, flash memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if instructions are transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. It should be understood, however, that computer-readable storage media and data storage media do not include connections, carrier waves, signals, or other transient media, but are instead directed to non-transient, tangible storage media. Disk and disc, as used, includes compact disc (CD), laser

disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc, where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

(101) Instructions may be executed by one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the term “processor”, as used may refer to any of the foregoing structure or any other structure suitable for implementation of the techniques described. In addition, in some aspects, the functionality described may be provided within dedicated hardware and/or software modules. Also, the techniques could be fully implemented in one or more circuits or logic elements.

(102) The techniques of this disclosure may be implemented in a wide variety of devices or apparatuses, including a wireless handset, an integrated circuit (IC) or a set of ICs (e.g., a chip set). Various components, modules, or units are described in this disclosure to emphasize functional aspects of devices configured to perform the disclosed techniques, but do not necessarily require realization by different hardware units. Rather, as described above, various units may be combined in a hardware unit or provided by a collection of interoperative hardware units, including one or more processors as described above, in conjunction with suitable software and/or firmware.

(103) It is to be recognized that depending on the example, certain acts or events of any of the methods described herein can be performed in a different sequence, may be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the method). Moreover, in certain examples, acts or events may be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors, rather than sequentially.

(104) In some examples, a computer-readable storage medium includes a non-transitory medium. The term “non-transitory” indicates, in some examples, that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium stores data that can, over time, change (e.g., in RAM or cache).

(105) Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A method of communicating using a breathing apparatus, the method comprising: receiving an audio signal from a sound acquisition unit; determining a state of the breathing apparatus based on the received audio signal, wherein the state is at least one of a first state and a second state; applying a first filter on the audio signal if the determined state is the first state, wherein the first filter has a first frequency response; applying a second filter on the audio signal if the determined state is the second state, wherein the second filter has a second frequency response different from the first frequency response of the first filter; generating an output signal based on the application of the first filter or the second filter on the audio signal; and a second computing device communicatively coupled to the article of PPE and comprising a second computer processor and a second memory; receiving the output signal at an output device; wherein the first state corresponds to a purge-off state and the second state corresponds to a purge-on state of the breathing apparatus.
2. The method of claim 1, wherein, in the purge-off state, a purge valve of the breathing apparatus is closed, wherein, in the purge-on state, the purge valve of the breathing apparatus is at least partially open, and wherein the purge valve fluidly communicates a facepiece of the breathing apparatus with an ambient when the purge valve is at least partially open.
3. The method of claim 1, wherein determining the state of the breathing apparatus further comprises: obtaining the audio signal over a history window, the history window corresponding to a first time period; dividing the history window into a plurality of time frames, each time frame

corresponding to a second time period less than the first time period; comparing an amplitude of the audio signal in the plurality of time frames with a predetermined noise threshold; and determining a purge noise frame count in the history window corresponding to a number of time frames in which the amplitude of the audio signal exceeds the predetermined noise threshold.

4. The method of claim 1, further comprising selecting the first filter from a set of first audio filters.

5. The method of claim 1, further comprising selecting the second filter from a set of second audio filters.

6. The method of claim 1, wherein the first frequency response of the first filter is inverse of a frequency response of a facepiece of the breathing apparatus.

7. The method of claim 1, wherein the first filter amplifies frequencies within a boost frequency range.

8. The method of claim 1, wherein the breathing apparatus is a self-contained breathing apparatus (SCBA) or a powered air-purifying respirator (PAPR).

9. A breathing apparatus comprising: a facepiece comprises a sound acquisition unit configured to generate an audio signal in response to a sound inside the facepiece; an audio processing unit configured to receive the audio signal from the sound acquisition unit, the audio processing unit configured to: determine a state of the breathing apparatus based on the received audio signal, wherein the state is at least one of a first state and a second state; apply a first filter on the audio signal if the determined state is the first state, wherein the first filter has a first frequency response; apply a second filter on the audio signal if the determined state is the second state, wherein the second filter has a second frequency response different from the first frequency response of the first filter; generate an output signal based on the application of the first filter or the second filter on the audio signal; and an output device receiving the output signal from the audio processing unit; a regulator mounted on the facepiece, the regulator comprising a purge valve that fluidly communicates the facepiece with an ambient when the purge valve is at least partially open, wherein the first state corresponds to a purge-off state when the purge valve is closed and the second state corresponds to a purge-on state when the purge valve is at least partially open.

10. The breathing apparatus of claim 9, wherein the audio processing unit is further configured to: obtain the audio signal over a history window, the history window corresponding to a first time period; divide the history window into a plurality of time frames, each time frame corresponding to a second time period less than the first time period; compare an amplitude of the audio signal in the plurality of time frames with a predetermined noise threshold; and determine a purge noise frame count in the history window corresponding to a number of time frames in which the amplitude of the audio signal exceeds the predetermined noise threshold.

11. The breathing apparatus of claim 9, wherein the audio processing unit is further configured to select the first filter from a set of first audio filters.

12. The breathing apparatus of claim 9, wherein the audio processing unit is further configured to select the second filter from a set of second audio filters.

13. The breathing apparatus of claim 9, wherein the first frequency response of the first filter is inverse of a frequency response of the facepiece.

14. The breathing apparatus of claim 9, wherein the first filter amplifies frequencies within a boost frequency range.

15. The breathing apparatus of claim 9, further comprising a wireless unit configured to wirelessly transmit the output signal from the audio processing unit to the output device.

16. The breathing apparatus of claim 9, wherein the breathing apparatus is a self-contained breathing apparatus (SCBA) or a powered air-purifying respirator (PAPR).

17. A method of communicating using a breathing apparatus having a facepiece and a purge valve, the method comprising: receiving an audio signal from a sound acquisition unit; determining a state of the breathing apparatus based on the received audio signal, wherein the state is at least one of a purge-off state and a purge-on state, wherein, in the purge-off-state, the purge valve is closed, and

wherein, in the purge-on state, the purge valve is at least partially open, the purge valve fluidly communicating the facepiece with an ambient when the purge valve is at least partially open; applying a first filter on the audio signal if the determined state is the purge-off state, wherein the first filter has a first frequency response; applying a second filter on the audio signal if the determined state is the purge-on state, wherein the second filter has a second frequency response different from the first frequency response of the first filter; generating an output signal based on the application of the first filter or the second filter on the audio signal; and receiving the output signal at an output device.

18. The method of claim 17, wherein determining the state of the breathing apparatus further comprises: obtaining the audio signal over a history window, the history window corresponding to a first time period; dividing the history window into a plurality of time frames, each time frame corresponding to a second time period less than the first time period; comparing an amplitude of the audio signal in the plurality of time frames with a predetermined noise threshold; and determining a purge noise frame count in the history window corresponding to a number of time frames in which the amplitude of the audio signal exceeds the predetermined noise threshold.
